

Class 15: Cloud computing

Montserrat (PID: A16536527)

2025-11-21

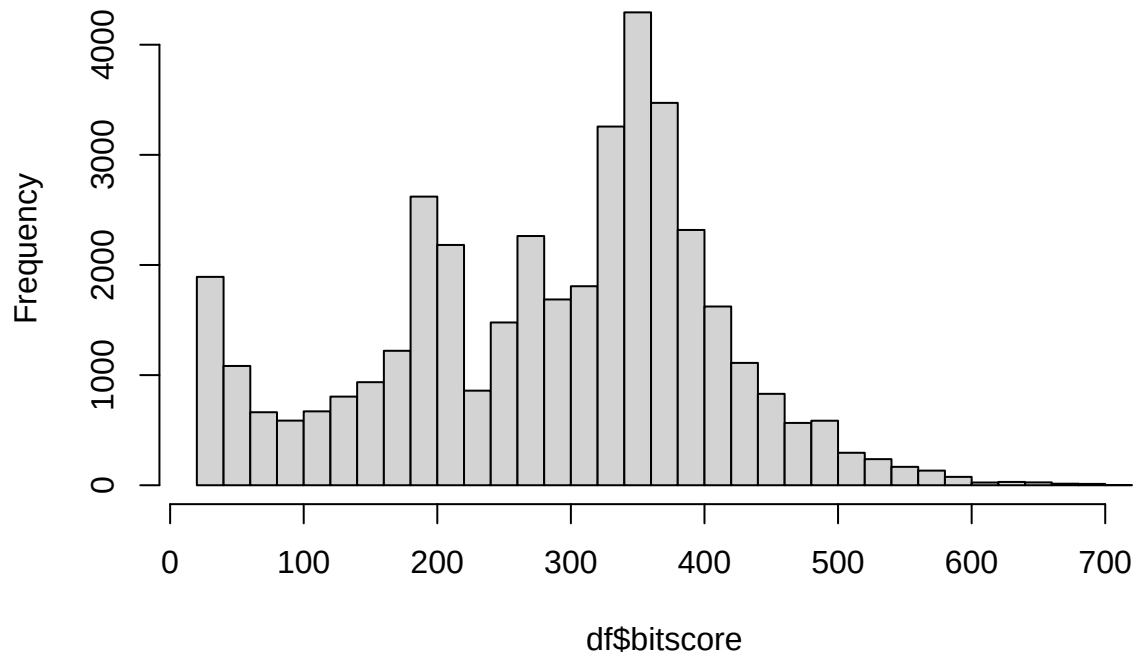
```
library(readr)
df <- read_tsv("mm-second.x.zebrafish.tsv")

## Rows: 39829 Columns: 12
## -- Column specification -----
## Delimiter: "\t"
## chr (2): NP_034603.2, XP_002663941.1
## dbl (10): 44.444, 90, 46, 1, 644, 733, 295, 380, 6.07e-20, 94.4
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
colnames(df) <- c("qseqid", "sseqid", "pident", "length", "mismatch", "gapopen", "qstart", "qend", "sstart", "send", "evalue", "bitscore")
```

Make a histogram of the \$bitscore values. You may want to set the optional breaks to be a larger number (e.g. breaks=30).

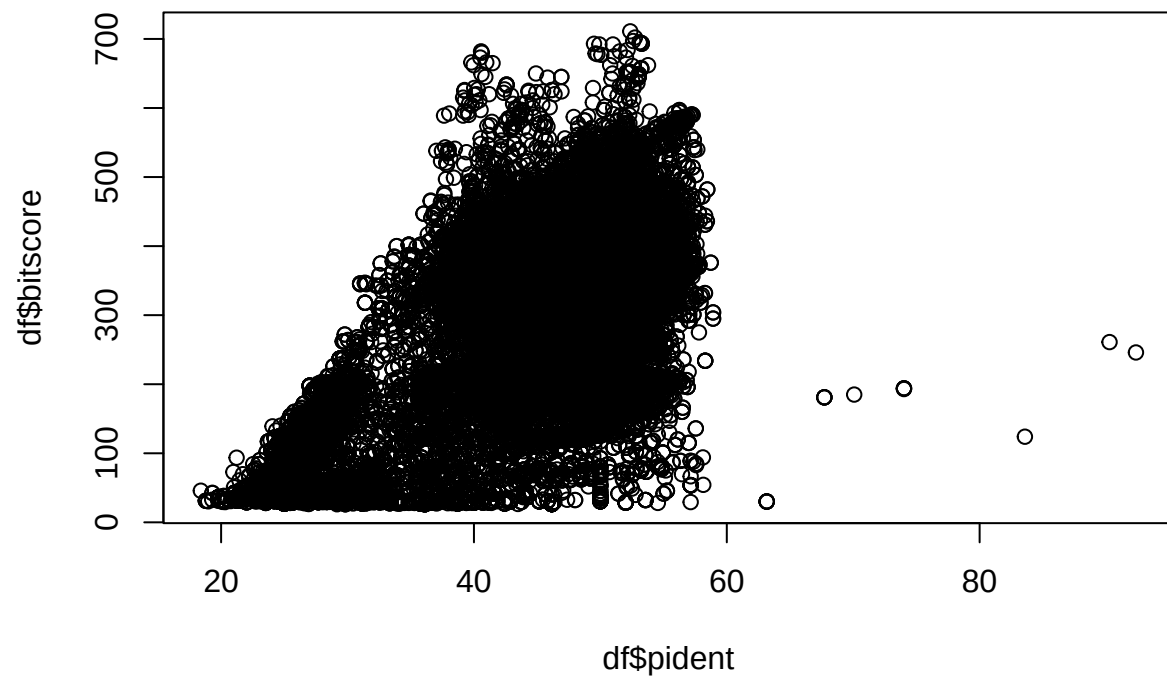
```
hist(df$bitscore, breaks=30)
```

Histogram of df\$bitscore

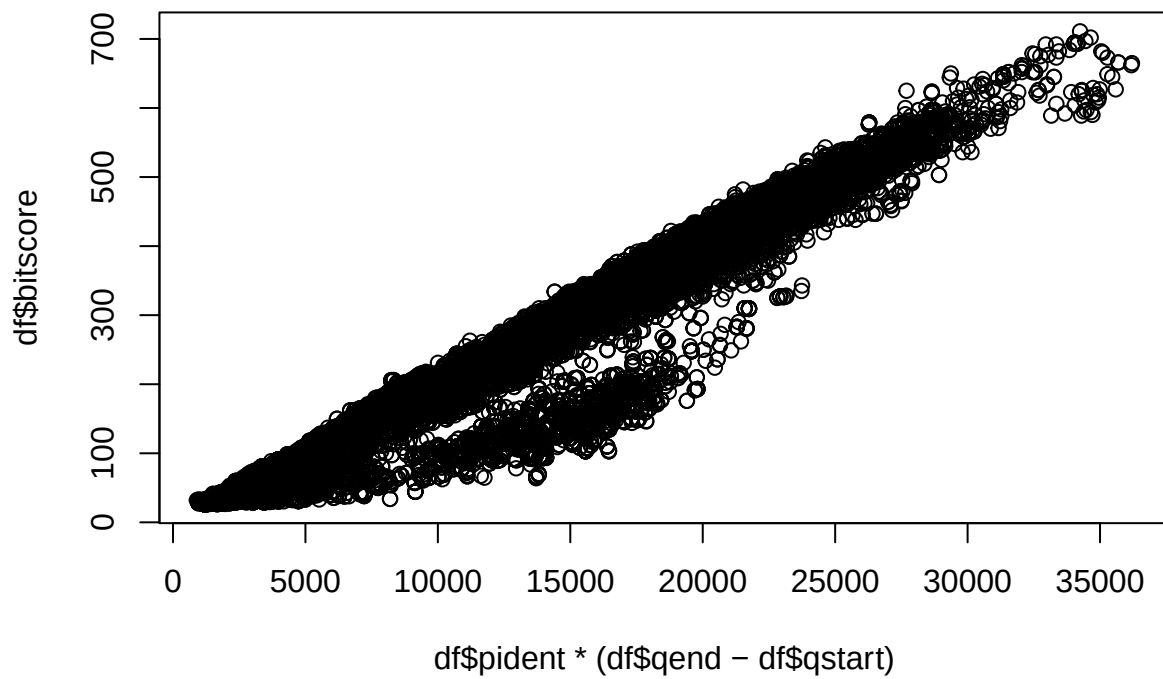


Is there a straightforward relationship between percent identity (*pident*) and *bitscore*(bitscore) for the alignments we generated?

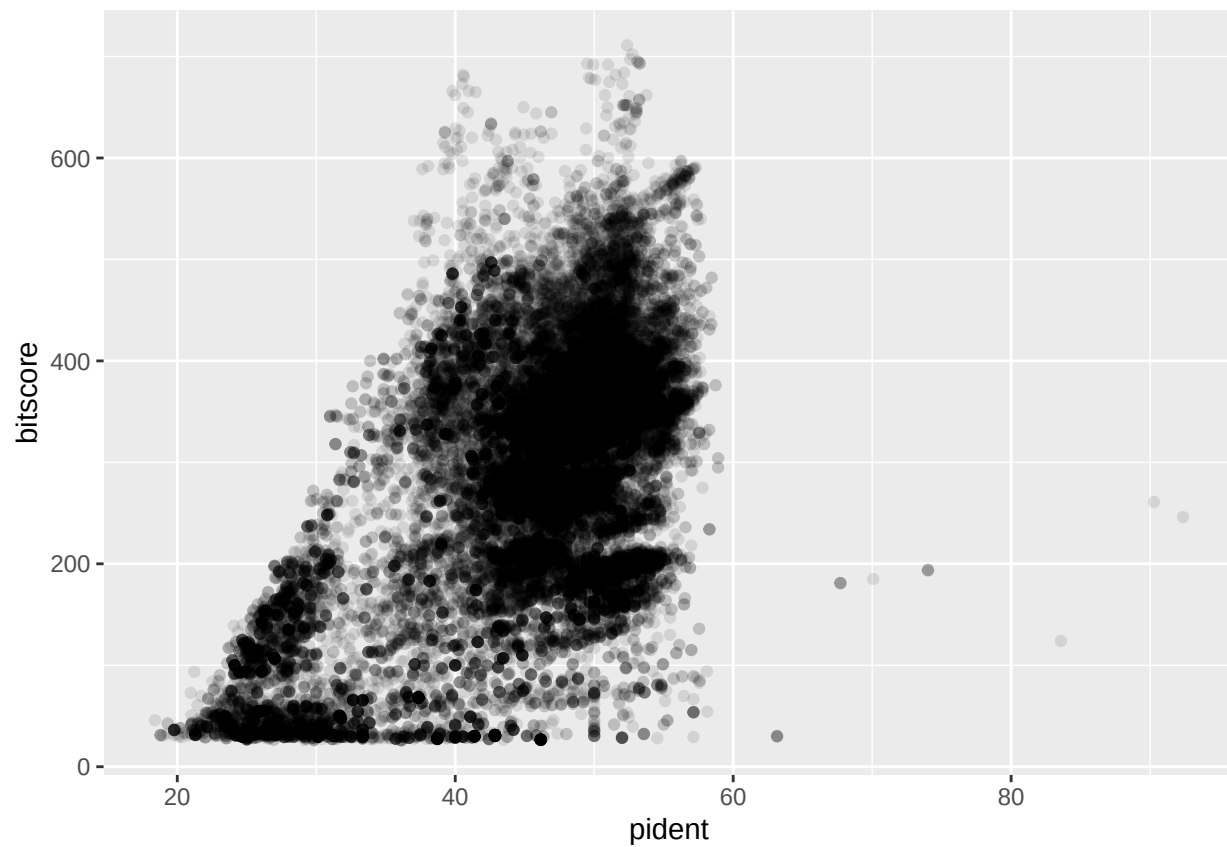
```
plot(df$pident,df$bitscore)
```



```
## Asuming your blast results are stored in an object called 'b'  
plot(df$pidet * (df$qend - df$qstart), df$bitscore)
```



```
library(ggplot2)
ggplot(df, aes(pident, bitscore)) + geom_point(alpha=0.1)
```



```
ggplot(df, aes((df$pident * (df$qend - df$qstart)), bitscore)) +  
  geom_point(alpha=0.1) + geom_smooth()
```

```
## `geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```

